

FixMatch-LS: Semi-supervised skin lesion classification with label smoothing

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ABSTRACT

Recent research on semi-supervised learning (SSL) is mainly based on the method of consistency regularization, which relies on data augmentation in the specific domain. Pseudolabelling is a more general method that has no such restrictions but is limited by noisy training. Medical datasets are a particular domain that exhibit a long-tail distribution. Combining these two limitations, we focus on the widespread use of weak augmentation to generate pseudolabels. We propose FixMatch-LS and a variant FixMatch-LS-v2 for medical image classification. First, we introduce label smoothing to change the pseudolabel threshold, which reduces the influence of noisy pseudolabels. In addition, pseudolabelling should be matched with consistency. A suitable consistency can constrain pseudolabelling to improve the quality of the pseudolabels. We validate our framework on skin lesion diagnoses from the ISIC 2018 and ISIC 2019 challenges, obtaining AUCs of 91.63%, 93.70%, 94.46%, and 95.44% on the four proportions of labelled data from ISIC 2018.

1. Introduction

With the help of a large number of labels, deep neural networks have achieved remarkable success in fully supervised models [6,7]. In the medical field, the annotation of high-quality data is time-consuming and requires a high level of proficiency from medical experts. Because labelling medical data requires professional medical knowledge, this process can only be done under the guidance of medical experts. Because unlabelled medical data are more readily available, SSL provides a method to improve classification performance by effectively utilizing the unlabelled data, aiming to reduce the labour of large-scale data annotation [8].

FixMatch [2] simplified SSL and obtained better classification performance by combining consistency regularization with pseudolabelling. For the same unlabelled image, FixMatch generated pseudolabels using weakly augmented samples and fed the strongly augmented samples into the model for training. Using SSL in the specific field of medicine usually requires dealing with unbalanced learning (due to high variability in disease prevalence; see Fig. 2 (a)). Unlike FixMatch, we

study the weakly augmented version (e.g., using only rotation, horizontal flip, and translation) of pseudolabelling, which is domain-independent of data augmentation. SRC-MT [1] uses only consistency regularization and the sample relation consistency paradigm to exploit the unlabelled samples of medical data. According to [34], pseudolabelling has no less effect than consistency. Based on SRC-MT, our research focuses on combining consistency regularization with a weakly augmented version of pseudolabelling.

The pseudolabelling process usually needs to consider the setting of the threshold [36]. Based on the SRC-MT experiment, the hard label makes the prediction probability of a large number of unlabelled data over 0.99 (see Appendix A Fig. 4 a), which is inconvenient for the selection of the pseudolabels. The performance of the model trained with a threshold of 0.95 performs poorly. Second, the annotation of medical data is a multifactor process, as in the hardware of medical equipment and the status of experts, so incorrect labelling may occur. Label smoothing is effective in alleviating the overfitting of data and reducing the interference of noisy image learning [5]. Therefore, we introduced label smoothing to mitigate these challenges.

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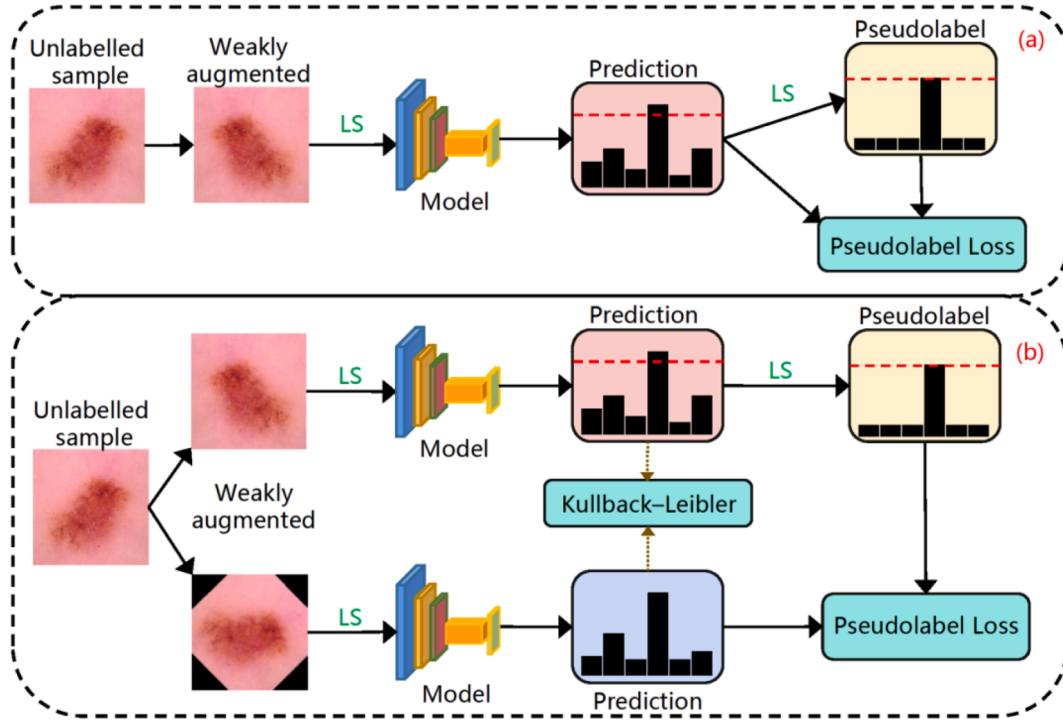
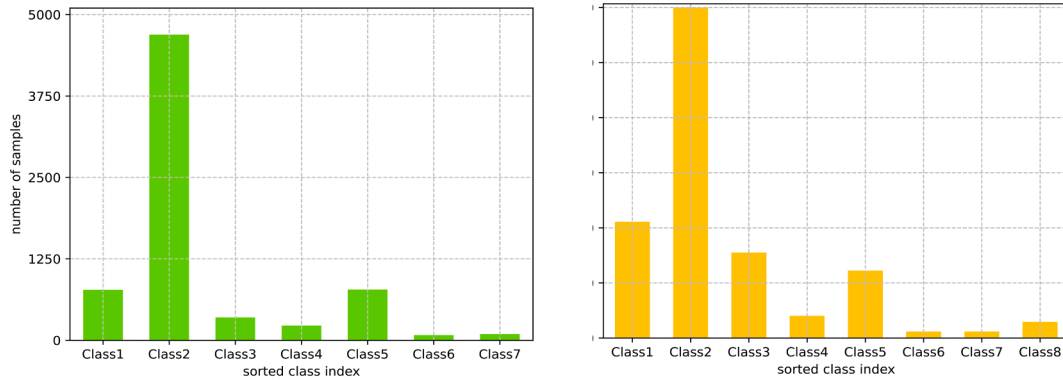
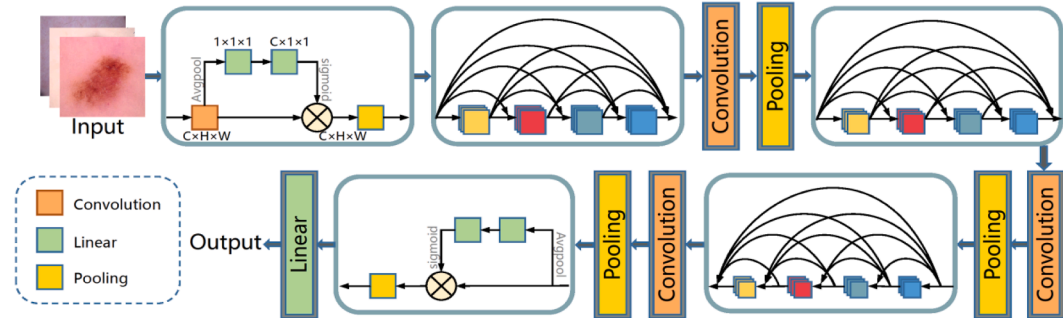


Fig. 1. Diagram of FixMatch-LS (a) and FixMatch-LS-v2 (b). An unlabelled sample is fed into the model to obtain predictions (red box). When the model assigns a probability to any class above a threshold (dotted line), the prediction is converted to a smoothed pseudolabel (yellow box). Then, Unlike FixMatch-LS, FixMatch-LS-v2 uses another prediction of the same sample (blue box) to calculate the pseudolabel loss.



(a) The imbalanced distribution on multiclass ISIC 2018 (left) and ISIC 2019 (right).



(b) Diagram of our DenseNet-121 model.

Fig. 2. In (a), we show the histograms of images per label for the multiclass ISIC 2018 [24] (left) and ISIC 2019 [24] (right). (b) Display diagram of our DenseNet-121 model. We adjusted the weight of the channel at the beginning and end. In the first block and the last block, leaky ReLU is between the two fully connected layers.

The FixMatch research [2] shows that the quality of pseudolabels is more important than the quantity in obtaining better performance. We found that consistency regularization can improve the quality of

pseudolabelling. We investigate the effect of two common consistency regularizations, mean square error (MSE) and Kullback–Leibler divergence (KL), on pseudolabelling at different labelled ratios. Different

consistencies have different effects on pseudolabels. We suggest that pseudolabelling should be matched with consistency.

Our method, named FixMatch-LS, uses label smoothing as its suffix to distinguish FixMatch. The diagram of our framework is shown in Fig. 1, and the model is shown in Fig. 2 (b). In comparison, FixMatch-LS (Fig. 1 (a)) can be considered a self-training approach. FixMatch-LS directly uses the prediction category output to both obtain pseudolabels and calculate the pseudolabel loss. FixMatch-LS-v2 (Fig. 1 (b)) produces more pseudolabels (more noisy labels) than FixMatch-LS because FixMatch-LS-v2 exploits the consistency between different augmentations of the same samples. While using MSE improves the quality of pseudolabels of FixMatch-LS, KL loss can obtain better pseudolabel quality than MSE loss. FixMatch-LS-v2 should be paired with KL loss.

Our contributions are as follows:

1. We propose a novel semi-supervised medical image classification framework that integrates four loss functions, including fully supervised classification loss, consistency regularization, the sample relation consistency paradigm (SRC), and pseudolabelled loss. The framework makes better use of unlabelled image features to improve classification performance.

2. Combining label smoothing with pseudolabelling, we alleviate the effect of noisy artificial labels with consistency regularization and relieve the imbalance of various categories of data by generating pseudolabels.

3. We observe the following:

- When training with fewer labelled samples, the model with KL loss performs better than the model with MSE loss.
- When training with more pseudolabels (and similarly with more noise labels), the model with KL loss performs better than the model with MSE loss, which is consistent with the results [49].
- Pseudolabelling needs to be matched with consistency. Inappropriate consistency will cause the model to underutilize the unlabelled data.
- The optimal pseudolabel threshold is 0.9 when the smoothing factor is typically 0.1. The sum of the two hyperparameters is 1.0. This means that we have the same confidence level for truth-valued labels as for pseudolabels. Unlike FixMatch, which takes the pseudolabel threshold to 0.95, it is just a hyperparameter.

4. We evaluate our method on the public medical skin lesion diagnosis datasets ISIC 2018 and ISIC 2019. FixMatch-LS-v2 obtained 91.63 % (4.02 % $\uparrow$), 93.70 % (3.39 % $\uparrow$), 94.46 % (0.88 % $\uparrow$) and 95.44 % (1.17 % $\uparrow$) AUCs on ISIC 2018 using 5 % (350 images), 10 %, 20 % and 30 % labelled data, respectively, and the other four indicators were superior.

We present consistency regularization, pseudolabelling, label smoothing, and some SSL work in medicine in the next section. In addition, we analyse FixMatch-LS and the differences between them. We specify the proposed framework in Section 3. Finally, we present the experimental results and an ablation study in Section 4.

2. Related work

Consistency regularization was first proposed by [9] and encouraged the model to produce the same output distribution after disturbance (data augmentation, dropout, or adversarial noise). For example, the Π model [10] uses the network outputs directly as the consistency targets. To be faster, the temporal ensembling (TE) method [10] integrates the unlabelled sample prediction of each epoch by exponential moving average (EMA) to obtain the consistency target. For more efficiency, the mean teacher (MT) [12] framework constructs a teacher model with the EMA weights of the student model. [43] proposed a model smoothing mechanism called the spatial ensemble that is complementary to EMA. Based on MT, VAT [13] proposed virtual adversarial disturbance to enhance the local smoothness of label distribution. UDA [14] shows that using strongly augmented samples can produce better results. FixMatch [2] expresses consistency through the strongly augmented sample and weakly augmented sample between the same samples. Noisy student

[15] focuses the disturbance mainly on the student model. CoMatch [47] combines graph-based contrastive loss with SSL, which generates better pseudolabels by embedding contrast learning and imposing a smoothing constraint on the classification probability.

The generality of self-training has been applied in many domains, including NLP [16], target detection [17], and image classification [15]. Pseudolabels and self-training [4] generate pseudolabels for unlabelled samples through the model trained on labelled samples. [36] extended the idea of pseudolabelling by incorporating confidence scores for unlabelled samples (i.e., obtaining the artificial label by filtering the model's prediction for unlabelled data with a constant confidence-based threshold). Aiming at a constant pseudolabel threshold of FixMatch, curriculum pseudolabelling (CPL) was proposed in FlexMatch [45]. FlexMatch uses the number of pseudolabels to represent the learning status and learning difficulty of different categories to flexibly adjust the pseudolabel threshold of different categories in each time step. PLACL [46] formulates the curriculum design problem as a decision-making problem and uses the reinforcement learning (RL) framework to solve it. While some studies have suggested that pseudolabelling is not competitive against other modern SSL algorithms on its own [20], recent SSL algorithms have combined pseudolabelling and consistency as a part of their pipeline to produce better results [2,18,19]. As mentioned above, pseudolabelling and consistency regularization have been used as components for many SSL techniques.

Label smoothing has been widely used successfully in a series of tasks in deep learning models, including image classification [21], speech recognition [22], and machine translation [23]. Since neural network training is sensitive to minimizing loss, minor changes such as label smoothing can effectively improve the performance of the model [21], improve generalization [25] and prevent the network from overfitting the data [3,26]. [5] pointed out that when distilling models from noisy data, label smoothing of the teacher is beneficial. [37] proposed instance-based label smoothing approaches in which the assigned smoothing factor is nonuniformly distributed along with the classes according to their similarity with the actual class. Label relaxation (LR) [51] changes the distribution of LS in a data-driven way, which reduces the risk of incorporating undesirable biases in the learning process.

SSL is a research topic that has attracted the attention of the medical community [28] due to the increase in data and expensive labelling costs [27]. [29] regularized the consistency of skin lesion segmentation. [30] proposed an attention residual model that classifies complicated lesions by self-attention ability. [31] proposed an active learning method to annotate the data. [38] applied a data augmentation technique based on generative adversarial networks (GANs) to skin lesion classification. Based on the feature matrix of samples, [1] proposed a sample relation consistency paradigm (SRC) to mine additional semantic information between different samples. [32] combined self-supervised mean-teacher pretraining and semi-supervised fine-tuning to achieve better performance on the chest X-ray14 dataset. [48] proposed anticurriculum pseudolabelling (ACPL) and obtained SOTA, which uses cross-distribution sample informativeness to select information-rich samples instead of category confidence thresholds.

Of the above work, our method is similar to FixMatch [2] and MixMatch [18]. They both combine consistency regularization and pseudolabelling. Unlike FixMatch, which embodies consistency by combining strongly and weakly augmented samples, we directly use consistency and the SRC to constrain the model to improve the quality of pseudolabelling and take advantage of original features from unlabelled data. FixMatch relies heavily on domain-specific data augmentation, while our framework uses weak augmentation without this limitation. MixMatch does not use pseudolabels but sharpened artificial labels to encourage the model to produce high-confidence predictions to minimize entropy [39]. We posit that the label smoothing and the threshold used in pseudolabelling in our framework have similar effects. Specifically, after adding label smoothing, since label smoothing encourages each example in the training set to be kept at the same distance from all

other classes of templates, the clusters are tighter and separated from each other [3]. This makes our framework with label smoothing more consistent with the assumptions underlying SSL [27]: low-density separation, cluster assumption, self-training assumption, etc. These similarities and differences suggest that our framework can be viewed as a weakly augmented version of FixMatch, where we have combined three techniques (label smoothing, pseudolabelling, and consistency regularization) while removing sharpening from MixMatch and strong augmentation (RandAugment [40], CTAugment [19]) from FixMatch. We explain the proposed framework in detail in the next section.

3. Method

3.1. Label smoothing

Consider a K-class classification task and its (one-hot encoded) hard label vector. For each vector, the position represents the index of the class, and vice versa. Soft targets are a weighted average of the hard targets and the uniform distribution over labels. The LS is expressed as follows:

$$y_i^{LS} = \begin{cases} 1 - \alpha, y_i = 1 \\ \frac{\alpha}{K-1}, y_i = 0 \end{cases} \quad (1)$$

where α is the smoothing factor and takes values between 0 and 1 (usually 0.1), y_i^{LS} is the soft label after smoothing of the hard label y_i , and K is the number of categories in the dataset. For example, with four classes and a smoothing factor $\alpha = 0.1$, the hard label $y = (0, 1, 0, 0)$ would be replaced by the soft label $y^{LS} = (0.033, 0.9, 0.033, 0.033)$. This procedure is generally understood as a means of regularization [33].

3.2. FixMatch-LS

Consider a K-class classification task; we denote a batch of a labelled dataset $X = \{(x_b, p_b) : b \in (1, \dots, B)\}$, where x_b is the training labelled data and p_b is their corresponding (one-hot encoded) hard label. Similarly, a batch of an unlabelled dataset is denoted $U = \{x_{ub} : u_b \in (1, \dots, u_B)\}$. Then, let $p_{\text{model}}(y|x)$ be the predicted class distribution produced by the model for input x , and let $H(p, q)$ be the cross-entropy between the two probability distributions p and q . Finally, let $LS(y)$ be the soft label of a hard label (one-hot encoded) y after label smoothing with α .

A. Overall Loss Function.

Our semi-supervised framework makes more effective use of the information in unlabelled images through consistent regularization and pseudolabelling methods. The total optimization objective function of FixMatch-LS and its variants is as follows.

$$L = L_s + \lambda_u L_u \text{ with } L_u = \lambda(\text{CLASS_WEIGHT} L_c + \beta L_{\text{src}}) + L_p \quad (2)$$

where L_s is the supervised loss; L_u is the unsupervised loss composed of the consistency regularization loss L_c , the sample relation consistency loss L_{src} and the pseudolabel loss L_p ; λ_u is the trade-off weight between the supervised and unsupervised losses; and λ is the trade-off weight between the consistency loss and pseudolabel loss. We will investigate the effect of this hyperparameter in ablation experiments D; CLASS_WEIGHT is a parameter used to balance categories in L_c , which is determined by dividing the total number of labelled data by the number of samples in each category; and β is a hyperparameter used to balance L_{src} and L_c .

B. Supervised Loss.

Minimizing the cross-entropy loss function encourages the predicted category distribution of labelled samples to be aligned with a hot truth label and similarly for unlabelled samples. We use label smoothing to change the truth labels of supervised data to affect the predicted class distribution of unlabelled samples. The loss function for supervised

learning is as follows:

$$L_s = \frac{1}{B} \sum_{b=1}^B H(\text{LS}(p_b), p_{\text{model}}(y|x_b)) \quad (3)$$

C. Consistency Regularization Loss and Sample Relation Consistency Paradigm.

Consistency regularization encourages the model to output similar prediction distributions by inputting a perturbed version of the same image [9]. We remove the Gaussian noise added in SRC-MT [1] because the model does not benefit from the smaller Gaussian noise. In UDA [14], replacing traditional noise injection methods with high-quality augmentation is recommended to improve consistency training. We base the difference between the weak augmentation function with different stochastics by the same sample to encourage the consistency of the output of the teacher model and the student model. The MSE consistency loss function is as follows:

$$L_c = L_{\text{MSE}} = \frac{1}{B + u_B} \sum_{i=1}^{B+u_B} \|p_{\text{model}}'(\text{LS}(y)|x_i') - p_{\text{model}}(\text{LS}(y)|x_i)\|_2^2 \quad (4)$$

p_{model}' is a teacher network, and p_{model} is a student network. x_i' and x_i are the same weakly augmented versions of the same sample. The SRC paradigm is as follows:

$$L_{\text{src}} = \frac{1}{B + u_B} \sum_{i=1}^{B+u_B} \|R^l(x_i') - R^l(x_i)\|_2^2 \quad (5)$$

where $R^l = \left[\frac{A^l \hat{A} \cdot (A^l)^T}{\|A^l \hat{A} \cdot (A^l)^T\|_2} \right]^T$ and A^l is the activation map of layer l. R^l is the relation matrix obtained by conducting L2 normalization for the inner product of the teacher feature map, and R^l is the student. For more detailed information about the SRC, see [1].

D. Pseudolabel Loss.

Training with limited data results in a trained classifier biased towards easy classes [34]. The training samples of the second category (a disease called melanocytic nevi) account for approximately 65 %, resulting in sample imbalance. The classifier is more inclined to the second category. Therefore, we filter all artificial labels of the second category to alleviate the extreme imbalance of the dataset. We first compute the student model's predicted class distribution given a weakly augmented sample of a given unlabelled image: $q_{ub} = p_{\text{model}}(y|x_{ub})$. Then, based on the output, FixMatch-LS assigns artificial labels to corresponding unlabelled samples whose confidence is higher than the threshold: $\hat{q}_{ub} = \arg \max(q_{ub})$. Finally, the smoothing loss of pseudolabels is calculated through the same output:

$$L_p = \frac{1}{u_B} \sum_{ub=1}^{u_B} 1(\max(q_{ub}) \geq T) H(\text{LS}(\hat{q}_{ub}), q_{ub}) \quad (6)$$

where $\hat{q}_{ub}$ is the pseudolabel after removing the second class and T is a scalar hyperparameter, which represents the threshold we assign to pseudolabels. We investigate the effect of this hyperparameter T in ablation experiments C.

3.3. FixMatch-LS-v2

For pseudolabelling, unlike FixMatch-LS, which is completely self-training, FixMatch-LS-v2 utilizes the consistency between weakly augmented data of the same samples. A diagram of FixMatch-LS-v2 is shown in Fig. 1 (b). Specifically, after obtaining the artificial labels $\hat{q}_{ub}$ from a given sample x_{ub} , we calculate the smoothing loss against the output of the model for x_{ub}' , which is a different version of the same weakly augmented sample:

Table 1

Comparison with semi-supervised learning methods on the ISIC 2018 dataset.

Method	Percentage		Metrics					
	Labelled	Unlabelled	AUC	Sensitivity	Specificity	Accuracy	F1	p-value
Upper Bound	100 %	0	95.83	79.15	93.86	95.02	73.05	
Self-training [42]	20 %	80 %	90.58	67.63	93.31	92.37	54.51	
SS-DCGAN [41]	20 %	80 %	91.28	67.72	92.56	92.27	54.10	
TCSE [29]	20 %	80 %	92.24	68.17	92.51	92.35	58.44	
TE [10]	20 %	80 %	92.70	69.81	92.55	92.26	59.33	
MT [12]	20 %	80 %	92.96	69.75	92.20	92.48	59.10	
SRG-MT [1]	20 %	80 %	93.58	71.47	92.72	92.54	60.68	
FlexMatch(RA) [45]	20 %	80 %	93.46 ± 0.13	71.25 ± 0.23	92.23 ± 0.17	93.40 ± 0.05	61.37 ± 0.22	p < 0.00
CoMatch(RA) [47]	20 %	80 %	91.10 ± 0.08	70.22 ± 0.25	92.29 ± 0.11	93.39 ± 0.04	61.94 ± 0.21	p < 0.00
FixMatch(RA) w/ SAW [52]	20 %	80 %	93.25 ± 0.14	75.43 ± 0.42	92.21 ± 0.08	93.19 ± 0.06	60.12 ± 0.23	
ACPL [48]	20 %	80 %	94.36	72.14			62.23	
FixMatch(RA) [2]	20 %	80 %	93.85 ± 0.17	72.91 ± 0.52	92.21 ± 0.03	93.23 ± 0.14	60.19 ± 0.31	0.005
FixMatch-LS (ours)	20 %	80 %	94.29 ± 0.05	74.15 ± 0.43	91.68 ± 0.12	93.22 ± 0.09	63.96 ± 0.46	0.02
FixMatch-LS w/ Mixup [18]	20 %	80 %	93.81 ± 0.11	72.82 ± 0.46	91.42 ± 0.14	93.05 ± 0.03	62.82 ± 0.09	p < 0.00
FixMatch-LS-v2 (ours)	20 %	80 %	94.49 ± 0.03	74.18 ± 0.43	91.94 ± 0.09	93.25 ± 0.05	64.15 ± 0.22	–

$$L_p = \frac{1}{u_B} \sum_{u_B=1}^{u_B} 1(\max(q_{u_B}) \geq T) H(LS(\hat{q}_{u_B}), p_{model}(y|x_{u_B})) \quad (7)$$

We investigate the effect of hyperparameter T in ablation experiment C. Compared to that of FixMatch-LS, the number of pseudolabels is doubled in this way, with a slight decrease in accuracy. To reduce the confirmation bias caused by the mislabelling of pseudolabels, we use KL loss as the consistency loss function.

$$L_c = L_{KL} = \frac{1}{B + \mu_B} \sum_{i=1}^{B + \mu_B} (p_{model}(LS(y)|x_i) \log \left(\frac{p_{model}(LS(y)|x_i)}{p_{model}(LS(y)|x_i)} \right)) \quad (8)$$

4. Experiment

4.1. Dataset setup

We performed our framework on the data from ISIC 2018 and ISIC 2019: Skin Lesion Analysis Towards Melanoma Detection [24,35]. The two datasets consisted of 10,015 and 25,531 skin lesion dermoscopy images with 7 and 8 categories, respectively. Due to the inconsistent size of ISIC 2019 raw images, we resized the images offline to 256 × 256 to facilitate model training. To compare with previous work, the entire dataset was randomly divided into 70 % for training, 10 % for validation, and 20 % for testing. As is typical in many SSL methods, we used data augmentation for both labelled and unlabelled data in the training

data, which is weak augmentation (e.g., using only rotation, horizontal flip, and translation). Models can benefit from data augmentation during validation and testing [11]; unlike the general testing process, we performed data augmentation on both validation and test set data, including vertical flipping and center crop, in addition to weak data augmentation using the training set. For the semi-supervised fine-tuning of the MT, we followed the reference [1] and the experiment with training sets containing different proportions of labelled data (5 % (350 images), 10 %, 20 %, and 30 %) for ISIC 2018. Since the ISIC 2019 dataset has two times more images than ISIC 2018, the percentage of labelled data division changed into 2 % (355 images), 5 %, 10 %, and 20 %. From the perspective of the application, we report classification results on the test set using five metrics, including AUC, accuracy, sensitivity, specificity, and F1.

4.2. Implementation details

We used the DenseNet-121 pretrained on ImageNet as our backbone and make slight adjustments according to Fig. 2 (b). We resized the original images to 224 × 224 for faster processing. We trained the network with adaptive momentum estimation (Adam) and set the learning rate to 0.0001. The batch size was set to 36, containing 9 labelled samples and 27 unlabelled samples in each mini-batch. Each training took 60 epochs, with the learning rate decaying by 0.1 at 6 and 12 epochs. The label smoothing was set to 0.1, and the pseudolabel confidence threshold was 0.9.

FixMatch [2] found that it is unnecessary to increase the weight term λ_u of unlabelled loss during training and let the pseudolabels add naturally. In our framework, we also did not increase the unlabelled loss of weight ($\lambda_u = 1$ in Eq.(2)). The generation of pseudolabels and the addition of unlabelled data were allowed at the same time. Furthermore, we applied the Gaussian warming-up function $\lambda(t) = 1 * e^{-(5(1-t/0)^2)}$ as λ , where θ is the total rising epoch. We set the EMA decay rate to 0.99, the ramp-up epoch θ to 30, and β to 1 in Eq.(2), according to [1]. The whole code was written in PyTorch. We used Nvidia RTX 2070 (8G) for training.

4.3. Experimental results on ISIC 2018

In this section, we show the semi-supervised experimental results of our framework on ISIC 2018. We experimented with FixMatch, FlexMatch and CoMatch on the ISIC 2018 dataset, setting the pseudolabel threshold T to 0.95, according to [2,45,46]. Table 1 shows the performance comparison of our framework with advanced methods with the setting of 20 % (1400) labelled data. The method based on self-training obtains higher specificity than the other methods, but its performance in

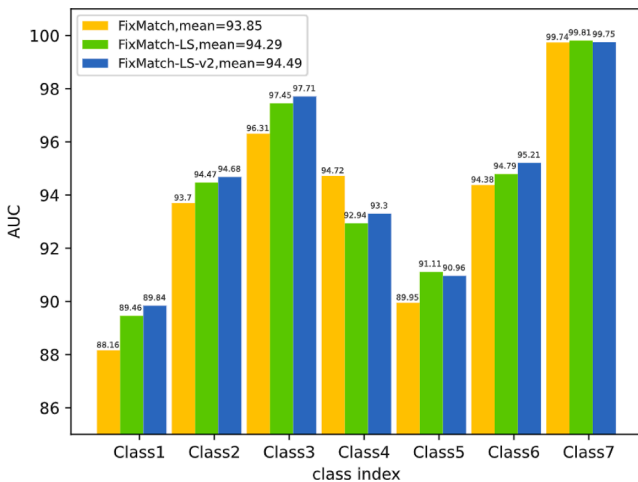


Fig. 3. Diagram of the AUC metrics of our method for each category on the ISIC 2018 dataset.

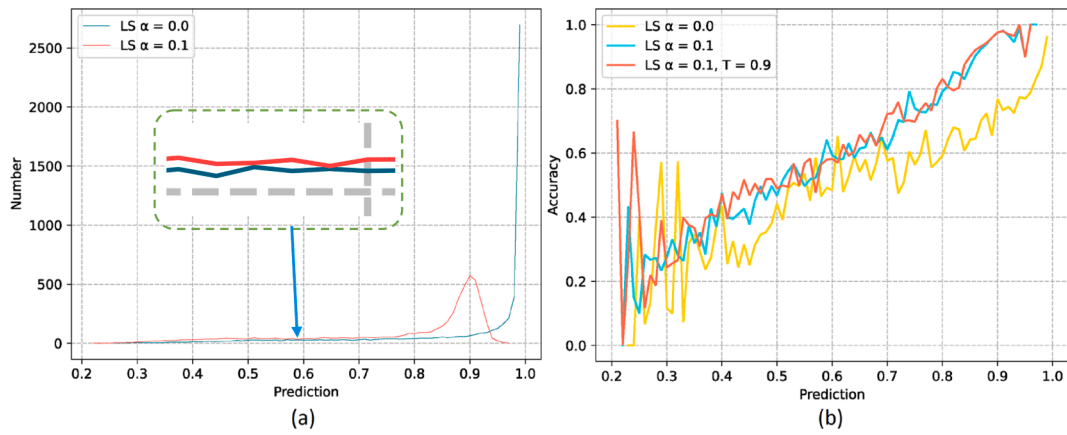


Fig. 4. Figure a (left) shows the distribution of the number of pseudolabels, and Figure b (right) shows their corresponding accuracy.

Table 2

Quantitative evaluation of our methods on the ISIC 2018 dataset with the different percentages of labelled data. * represents training with MSE loss consistency rather than KL loss.

Method	Percentage		Metrics				
	Labelled	Unlabelled	AUC	Sensitivity	Specificity	Accuracy	F1
Upper Bound ($\alpha = 0.1$)	100 %	0	95.83	79.15	93.86	95.02	73.05
Upper Bound ($\alpha = 0.0$)	100 %	0	95.73	78.02	94.17	95.11	73.03
SRC-MT [1]	5 %	95 %	87.61	62.04	89.36	88.77	46.26
FixMatch [2]	5 %	95 %	86.05 $\pm$ 0.17	55.69 $\pm$ 0.57	89.43 $\pm$ 0.20	89.70 $\pm$ 0.30	46.29 $\pm$ 0.31
FixMatch-LS (ours)	5 %	95 %	91.36 $\pm$ 0.02	64.39 $\pm$ 0.60	88.97 $\pm$ 0.14	90.61 $\pm$ 0.04	53.24 $\pm$ 0.19
FixMatch-LS-v2 (ours)	5 %	95 %	91.64 $\pm$ 0.04	64.74 $\pm$ 0.57	89.64 $\pm$ 0.09	90.77 $\pm$ 0.06	53.71 $\pm$ 0.33
SRC-MT [1]	10 %	90 %	90.31	66.29	90.47	89.30	50.02
FixMatch [2]	10 %	90 %	91.72 $\pm$ 0.05	71.15 $\pm$ 0.59	89.22 $\pm$ 0.17	91.59 $\pm$ 0.12	54.20 $\pm$ 0.49
FixMatch-LS (ours)	10 %	90 %	93.43 $\pm$ 0.10	73.53 $\pm$ 0.33	89.87 $\pm$ 0.14	91.53 $\pm$ 0.14	58.70 $\pm$ 0.26
FixMatch-LS-v2 (ours)	10 %	90 %	93.63 $\pm$ 0.08	73.74 $\pm$ 0.47	90.22 $\pm$ 0.21	91.93 $\pm$ 0.16	59.40 $\pm$ 0.32
SRC-MT [1]	20 %	80 %	93.58	71.47	92.72	92.54	60.68
FixMatch [2]	20 %	80 %	93.85 $\pm$ 0.17	72.91 $\pm$ 0.52	92.21 $\pm$ 0.03	93.23 $\pm$ 0.14	60.19 $\pm$ 0.31
FixMatch-LS (ours)*	20 %	80 %	94.29 $\pm$ 0.05	74.15 $\pm$ 0.43	91.68 $\pm$ 0.12	93.22 $\pm$ 0.09	63.96 $\pm$ 0.46
FixMatch-LS-v2 (ours)	20 %	80 %	94.49 $\pm$ 0.03	74.18 $\pm$ 0.43	91.94 $\pm$ 0.09	93.25 $\pm$ 0.05	64.15 $\pm$ 0.22
SRC-MT [1]	30 %	70 %	94.27	74.59	92.82	93.11	63.54
FixMatch [2]	30 %	70 %	94.34 $\pm$ 0.08	74.88 $\pm$ 0.54	92.63 $\pm$ 0.11	93.81 $\pm$ 0.04	63.41 $\pm$ 0.17
FixMatch-LS (ours)*	30 %	70 %	95.14 $\pm$ 0.07	76.22 $\pm$ 0.21	92.62 $\pm$ 0.13	93.62 $\pm$ 0.07	64.83 $\pm$ 0.30
FixMatch-LS-v2 (ours)	30 %	70 %	95.42 $\pm$ 0.05	77.64 $\pm$ 0.56	92.78 $\pm$ 0.12	93.68 $\pm$ 0.09	65.33 $\pm$ 0.37

terms of other indicators is limited. Our framework outperforms FixMatch by 2.8 % in sensitivity and F1 and does not differ significantly in other metrics. After adding SAW, the sensitivity of FixMatch is

significantly improved, obtaining a performance result of 75.43 %, but no significant improvement in other metrics. FixMatch-LS-v2 was 1.24 % and 4.03 % higher than the method in terms of AUC and F1 values.

Table 3

Quantitative evaluation of our methods on the ISIC 2019 dataset with the different percentages of labelled data. * represents training with MSE loss consistency rather than KL loss.

Method	Percentage		Metrics				
	Labelled	Unlabelled	AUC	Sensitivity	Specificity	Accuracy	F1
Upper Bound	100 %	0	97.21	84.37	93.40	94.33	71.78
SRC-MT [1]	2 %	98 %	82.92 $\pm$ 0.07	54.35 $\pm$ 0.28	86.79 $\pm$ 0.03	87.49 $\pm$ 0.04	40.45 $\pm$ 0.21
FixMatch [2]	2 %	98 %	84.25 $\pm$ 0.09	55.49 $\pm$ 0.75	84.96 $\pm$ 0.37	86.13 $\pm$ 0.36	38.37 $\pm$ 0.25
FixMatch-LS (ours)	2 %	98 %	86.49 $\pm$ 0.05	57.06 $\pm$ 0.26	86.05 $\pm$ 0.06	87.42 $\pm$ 0.08	41.89 $\pm$ 0.14
FixMatch-LS-v2 (ours)	2 %	98 %	86.42 $\pm$ 0.07	57.44 $\pm$ 0.40	86.33 $\pm$ 0.07	87.72 $\pm$ 0.08	42.74 $\pm$ 0.24
SRC-MT [1]	5 %	95 %	86.19 $\pm$ 0.41	56.47 $\pm$ 0.49	87.95 $\pm$ 0.15	89.06 $\pm$ 0.05	45.78 $\pm$ 0.08
FixMatch [2]	5 %	95 %	87.82 $\pm$ 0.10	60.19 $\pm$ 0.11	88.27 $\pm$ 0.18	89.27 $\pm$ 0.09	46.25 $\pm$ 0.15
FixMatch-LS (ours)	5 %	95 %	89.21 $\pm$ 0.06	59.93 $\pm$ 0.32	87.83 $\pm$ 0.04	89.17 $\pm$ 0.07	46.94 $\pm$ 0.21
FixMatch-LS-v2 (ours)	5 %	95 %	89.19 $\pm$ 0.06	60.38 $\pm$ 0.30	88.48 $\pm$ 0.02	89.43 $\pm$ 0.05	48.00 $\pm$ 0.16
SRC-MT [1]	10 %	90 %	89.17 $\pm$ 0.02	66.27 $\pm$ 0.13	88.19 $\pm$ 0.05	89.34 $\pm$ 0.03	50.26 $\pm$ 0.13
FixMatch [2]	10 %	90 %	91.33 $\pm$ 0.08	68.08 $\pm$ 0.67	89.51 $\pm$ 0.17	90.15 $\pm$ 0.15	50.71 $\pm$ 0.28
FixMatch-LS (ours)	10 %	90 %	92.13 $\pm$ 0.10	69.43 $\pm$ 0.49	88.41 $\pm$ 0.11	89.89 $\pm$ 0.09	53.90 $\pm$ 0.36
FixMatch-LS-v2 (ours)	10 %	90 %	92.29 $\pm$ 0.14	69.45 $\pm$ 0.39	89.08 $\pm$ 0.14	90.14 $\pm$ 0.05	53.95 $\pm$ 0.39
SRC-MT [1]	20 %	80 %	91.18 $\pm$ 0.07	70.87 $\pm$ 0.23	90.35 $\pm$ 0.08	90.92 $\pm$ 0.05	55.69 $\pm$ 0.22
FixMatch [2]	20 %	80 %	93.43 $\pm$ 0.05	69.65 $\pm$ 0.11	91.29 $\pm$ 0.12	91.86 $\pm$ 0.10	58.18 $\pm$ 0.30
FixMatch-LS (ours)*	20 %	80 %	93.45 $\pm$ 0.09	71.22 $\pm$ 0.30	90.32 $\pm$ 0.06	91.44 $\pm$ 0.07	59.45 $\pm$ 0.21
FixMatch-LS-v2 (ours)	20 %	80 %	93.84 $\pm$ 0.10	71.32 $\pm$ 0.34	90.84 $\pm$ 0.18	91.81 $\pm$ 0.09	59.47 $\pm$ 0.28

FlexMatch[45] performs worse than FixMatch on unbalanced datasets. FlexMatch does not suit the long-tail scenario where unlabelled data belonging to each class are extremely unbalanced. Notable, our framework improves the sensitivity and F1 by almost 2 % compared to the latest method ACPL [48] with the same AUC. In our framework, the difference between FixMatch-LS and FixMatch-LS-v2 in 20 % of labelled data is small. Applying the mixup augmentation [18] (mixing factor of 0.75) to the unlabelled data, we applied FixMatch-LS to the mixed data and obtained worse performance than with the original. We speculated that the combination of mixup and label smoothing does not apply to the classification of dermoscopy images of skin lesions. The AUC performance results for each category are shown in Fig. 3. The AUC improved in five of the seven diseases, with no significant difference between the three methods in category 7.

Table 2 shows the influence of different percentages of labelled data on the semi-supervised learning method. The experimental results show that FixMatch-LS-v2 is slightly better than FixMatch-LS in our framework. For each scale, with similar specificity, our framework shows significant improvement compared to SRC-MT in terms of the other 4 metrics, with a minimum improvement of 0.57 % as well as a maximum gain of 9.38 %. Our method is also competitive enough compared to FixMatch. FixMatch-LS-v2 has an average 2.25 % improvement compared to FixMatch. In particular, our framework has a higher sensitivity (i.e., low underdiagnosis rate) and F1 when only 5 % (350) of the labelled data are available. In our framework, the increment of specificity relative to sensitivity is small, and sensitivity itself is approximately 20 % lower than specificity, which is consistent with the constrained relationship between the two metrics. In addition, an AUC of 95.42 % is obtained on 30 % of the labelled data, and the sensitivity reaches 77.64 %, which is closer to the upper bound performance of the 100 % fully labelled model. When using 100 % of the labelled samples, our fully supervised performance AUC is 95.83 %, compared to 95.43 % for SRC-MT. These experimental results demonstrate the effectiveness of our efforts to leverage unlabelled data more fully in the SSL problem. For more information on the selection of consistency and the effect on pseudolabelling, please refer to the ablation study.

4.4. Experimental results on ISIC 2019

ISIC 2019 is a more unbalanced dataset than ISIC 2018 (see Fig. 2 (a)). Table 3 shows the impact of the four semi-supervised methods with different percentages of labelled data. The fully supervised model with 100 % (17732) labelled data exhibits the upper bound performance. Our method significantly outperforms SRC-MT and outperforms FixMatch in most cases. FixMatch-LS-v2 is improved compared to FixMatch-LS. In particular, our framework performed better with only 2 % (355) of the labelled data. Specifically, using only 2 % of labelled data, our framework gained more than 2 % improvement in terms of AUC, sensitivity and F1 compared to other methods. All four metrics of FixMatch are lower than those of SRC-MT at the same scale, which is similar to the results obtained on ISIC 2018. FixMatch obtained the highest accuracy of 91.86 % and specificity of 91.29 % when using 20 % (3548) of labelled data. The possible reason for the above phenomenon is that FixMatch [2] can simulate real images by strong augmentation, such as colour, brightness, and contrast, based on a certain amount of labelled data (e.g., more than 700–900 images), which improves the generalization and robustness of the model while increasing the sample diversity. When there are only a small number of labelled samples (350 images), the perturbations generated by strong augmentation may cause a collapse in predictive diversity [44]. At this point, our framework can achieve relatively better performance by weak augmentation with label smoothing.

4.5. Ablation study

A. Impact of each component of our framework.

Table 4

Ablation experiments on the different pretrained CNN models as backbone networks of FixMatch-LS-v2 based on the 20% truth value label.

BackBone	Metrics				
	AUC	Sensitivity	Specificity	Accuracy	F1
DenseNet-121	94.49 ± 0.03	74.18 ± 0.43	91.94 ± 0.09	93.25 ± 0.05	64.15 ± 0.22
MobileNet	92.73 ± 0.46	68.78 ± 0.71	91.07 ± 0.16	92.52 ± 0.18	58.52 ± 0.41
VGG-11	92.52 ± 0.18	67.06 ± 0.59	91.35 ± 0.17	92.62 ± 0.09	59.29 ± 0.33
VGG-16	92.41 ± 0.32	64.70 ± 0.51	91.74 ± 0.20	92.93 ± 0.10	57.62 ± 0.30
ResNet-18	92.72 ± 0.35	70.05 ± 0.54	90.46 ± 0.18	92.25 ± 0.11	59.94 ± 0.76
ResNet-34	92.28 ± 0.21	68.10 ± 0.77	91.36 ± 0.12	92.60 ± 0.13	60.08 ± 0.38
ResNet-50	93.30 ± 0.11	69.62 ± 0.47	91.71 ± 0.21	93.00 ± 0.13	62.51 ± 0.43

Table 5

Ablation experiments on each component of FixMatch-LS-v2 based on the 5% truth value label.

Ablation	Metrics				
	AUC	Sensitivity	Specificity	Accuracy	F1
FixMatch-LS-v2	88.36	59.11 ±	90.36 ±	89.97 ±	51.05
w/o LS ($\alpha = 0.0$, $T = 0.95$)	± 0.40	0.65	0.10	0.16	± 0.45
FixMatch-LS-v2	91.26	62.46 ±	88.86 ±	90.83 ±	52.63
w/o test data augment	± 0.29	0.60	0.05	0.18	± 0.19
FixMatch-LS-v2	91.35	63.51 ±	88.97 ±	90.30 ±	52.92
w/o pseudolabelling ($\alpha = 0.1$, $T = 1.0$)	± 0.10	0.18	0.20	0.08	± 0.09
FixMatch-LS-v2	91.02	62.25 ±	88.77 ±	90.13 ±	50.80
w/ DenseNet-121 backbone	± 0.05	1.18	0.10	0.25	± 0.65
FixMatch-LS-v2	90.11	61.81 ±	89.84 ±	90.60 ±	51.86
w/o KL and SRC	± 0.20	0.41	0.12	0.10	± 0.28
FixMatch-LS-v2	90.17	61.36 ±	89.84 ±	90.33 ±	51.82
w/o KL	± 0.26	0.37	0.29	0.23	± 0.32
FixMatch-LS-v2	91.33	63.44 ±	89.38 ±	90.72 ±	53.05
w/o SRC	± 0.04	0.29	0.08	0.17	± 0.36
FixMatch-LS-v2	89.77	61.64 ±	90.01 ±	90.17 ±	52.54
w/o CLASS WEIGHT	± 0.32	0.91	0.15	0.38	± 0.51
FixMatch-LS-v2	91.05	63.61 ±	89.35 ±	90.52 ±	52.82
w/ category 2	± 0.21	0.36	0.10	0.14	± 0.61
FixMatch-LS-v2	90.89	64.67 ±	89.08 ±	90.13 ±	52.03
w/ Gaussian noise	± 0.32	0.20	0.14	0.13	± 0.44
FixMatch-LS-v2 ($\alpha = 0.1$, $T = 0.9$)	91.64 ± 0.04	64.74 ± 0.57	89.64 ± 0.09	90.77 ± 0.06	53.71 ± 0.33

In our proposed method, the choice of backbone is also a crucial issue. We tested several standard CNN networks (MobileNet, VGG, ResNet, and DenseNet) and selected the best performing network. Table 4 shows the results of choosing the backbone network. DenseNet-121 offers the best performance as a backbone network. Since the framework combines various components, we investigated the effect of removing or adding components to provide additional insight into what makes our framework performant. Specifically, we carried out the ablation experiment on 350 (5 %) labelled images of ISIC 2018; the results of FixMatch-LS-v2 are shown in Table 5. Each component contributed to the performance of FixMatch-LS-v2. The most significant overall improvement in the model is seen with the addition of label smoothing. Directly adding pseudolabels leads to performance degradation, while label smoothing significantly reduces the impact of

Table 6

All experiments are based on ISIC 2018. The underlined part is the parameter chosen for our experiments.

Labelled data	Model	Consistency	Metrics					
			AUC	Sensitivity	Specificity	Accuracy	F1	PL avg acc
5 %	FixMatch-LS		90.05 ± 0.06	62.77 ± 0.35	89.51 ± 0.08	90.39 ± 0.12	51.88 ± 0.30	65.1
	FixMatch-LS	MSE + SRC	90.78 ± 0.24	60.75 ± 0.57	89.12 ± 0.16	90.60 ± 0.15	53.21 ± 0.14	72.4
	FixMatch-LS	KL + SRC	91.36 ± 0.02	64.39 ± 0.60	88.97 ± 0.14	90.61 ± 0.04	53.24 ± 0.19	72.9
	FixMatch-LS-v2		90.30 ± 0.08	61.65 ± 0.17	89.63 ± 0.11	90.25 ± 0.12	51.57 ± 0.21	57.0
	FixMatch-LS-v2	MSE + SRC	91.05 ± 0.16	60.62 ± 0.34	89.61 ± 0.11	90.73 ± 0.29	53.72 ± 0.19	67.9
	<u>FixMatch-LS-v2</u>	<u>KL + SRC</u>	91.64 ± 0.04	64.74 ± 0.57	89.64 ± 0.09	90.77 ± 0.06	53.71 ± 0.33	71.3
	FixMatch-LS		93.90 ± 0.09	73.58 ± 0.71	91.70 ± 0.13	92.74 ± 0.18	62.16 ± 0.54	84.0
	FixMatch-LS	MSE + SRC	94.29 ± 0.05	74.15 ± 0.43	91.68 ± 0.12	93.22 ± 0.09	63.96 ± 0.46	88.8
20 %	FixMatch-LS	KL + SRC	94.55 ± 0.12	74.73 ± 0.28	91.35 ± 0.12	92.90 ± 0.07	62.47 ± 0.31	89.1
	FixMatch-LS-v2		93.58 ± 0.14	72.93 ± 0.40	92.25 ± 0.07	92.81 ± 0.14	62.75 ± 0.36	80.2
	FixMatch-LS-v2	MSE + SRC	94.17 ± 0.18	73.89 ± 0.48	91.87 ± 0.10	93.08 ± 0.12	63.52 ± 0.35	85.5
	<u>FixMatch-LS-v2</u>	<u>KL + SRC</u>	<u>94.49 ± 0.03</u>	<u>74.18 ± 0.43</u>	<u>91.94 ± 0.09</u>	<u>93.25 ± 0.05</u>	<u>64.15 ± 0.22</u>	<u>89.4</u>

mislabels. The use of data augmentation in the test set improves the overall accuracy. Modifying the weight of the channel in the network results in a significant improvement. The addition of consistent regularization has a significant impact on model performance. The parameter CLASS_WEIGHT in consistency regularization plays a key role in alleviating the problem of category imbalance. Whether the second class in the pseudolabel is removed or not has little effect. Adding Gaussian noise does not bring positive benefits. This may be caused by extreme imbalanced data and inconsistency with the Gaussian distribution.

B. Impact of consistency and pseudolabelling.

Ablation experiments were performed on four proportions of labelled data from each of the two datasets. Table 6 shows the impact of consistency and pseudolabels on the model and the effect of consistency regularization on pseudolabelling. We added an indicator pseudolabel average accuracy (PL avg acc) to illustrate the influence of consistency regularization on pseudolabels. Note that the overall number of pseudolabels is smaller and the change in the PL avg acc is relatively larger because we removed all pseudolabels for which the pseudolabel was the second class (the one with the highest number) since the inclusion of that category would have led to more extreme imbalance.

ISIC 2019 is an 8-category dataset, only one more than ISIC 2018. However, the sample of the newly added categories includes data that belong to the tail of the long-tail distribution. First, Table 2 shows that FixMatch-LS uses MSE loss on 20 % (1400) and 30 % (2100) of labelled data from the ISIC 2018 dataset, while Table 3 shows that it only uses MSE loss on 20 % (3548) of labelled data from ISIC 2019. Second, as shown in Table 6, between the two proportions, the model uses 5 % of the labelled data more effectively than 20 % of the KL loss than the MSE loss. Within two proportions of labelled data, the difference between using KL loss or MSE loss is small. Third, the accuracy of pseudolabels trained with 5 % of the labelled data is lower than that of pseudolabels obtained by training with 20 % of the labelled data. The former is

affected by the consistency variation more than the latter. Fourth, FixMatch-LS-v2 can obtain better performance using KL loss than MSE loss.

Based on the experimental results, we speculate that the consistency strength of KL loss is higher than that of MSE loss. Consistency regularization controls the trade-off between the quality and the quantity of pseudolabels. Consistency here includes consistency between samples and added to the samples. The accuracy of pseudolabels increases with the strength of consistency regularization, while the amount of pseudolabels contributing to L_p in Eq. (7) decreases. This suggests that when the model performance is insufficient, i.e., there are fewer labelled samples, a higher intensity of consistency regularization is needed to constrain the semi-supervised training process. When we exploit the consistency among samples, stronger consistency regularization is needed to constrain the process to obtain high-quality pseudolabels.

C. Impact of the pseudolabelling threshold T.

We investigated the different hyperparameter settings for the pseudolabelling threshold T in Eq. (6) and Eq. (7). The threshold value controls the trade-off between the quality and quantity of pseudolabels [2]. The accuracy of pseudolabels for unlabelled data increases with increasing threshold values, while the amount of unlabelled data decreases. Specifically, the pseudolabelling method is sensitive to the threshold T, which changes every 0.01. We select six values of each method for presentation in Table 7. Note that FixMatch-LS is relatively insensitive to the threshold because label smoothing makes the prediction distribution of unlabelled images more uniform, and the number of pseudolabels between each probability is relatively small. FixMatch-LS-v2 makes the number of pseudolabels for each threshold at least double because of the consistency applied. The variation in the threshold value makes pseudolabelling more incorrect and causes degradation in model performance. The results of our experiments recommend the optimal value of T to be 0.9. Notably, $\alpha = 0.1$ can be interpreted as the model

Table 7

Quantitative evaluation of our methods with different thresholds T. Experiments were performed based on 20 % of the labelled data with label smoothing $\alpha = 0.1$. The underlined part is the parameter chosen for our experiments.

Method	Hyperparameter threshold T	Metrics					
		AUC	Sensitivity	Specificity	Accuracy	F1	
FixMatch-LS	0.87	94.06 ± 0.10	73.86 ± 0.38	91.85 ± 0.08	93.19 ± 0.05	64.11 ± 0.31	
	0.88	94.15 ± 0.11	73.78 ± 0.38	91.64 ± 0.10	93.19 ± 0.05	64.34 ± 0.29	
	0.89	94.27 ± 0.05	73.99 ± 0.24	91.73 ± 0.13	93.28 ± 0.04	64.13 ± 0.18	
	<u>0.90</u>	94.29 ± 0.05	74.15 ± 0.43	<u>91.68 ± 0.12</u>	<u>93.22 ± 0.09</u>	<u>63.96 ± 0.46</u>	
	0.91	94.13 ± 0.05	73.92 ± 0.33	91.52 ± 0.04	93.15 ± 0.11	64.38 ± 0.26	
	0.92	94.16 ± 0.05	74.11 ± 0.40	91.62 ± 0.11	93.14 ± 0.10	63.89 ± 0.49	
FixMatch-LS-v2	0.87	94.28 ± 0.16	71.56 ± 0.53	92.07 ± 0.20	92.81 ± 0.11	62.07 ± 0.39	
	0.88	93.74 ± 0.11	73.22 ± 0.71	91.94 ± 0.22	92.53 ± 0.23	61.51 ± 0.52	
	0.89	94.05 ± 0.29	74.29 ± 0.52	91.60 ± 0.16	92.63 ± 0.12	61.77 ± 0.37	
	<u>0.90</u>	<u>94.49 ± 0.03</u>	<u>74.18 ± 0.43</u>	<u>91.94 ± 0.09</u>	93.25 ± 0.05	64.15 ± 0.22	
	0.91	94.22 ± 0.11	73.44 ± 0.36	91.74 ± 0.15	93.09 ± 0.07	63.69 ± 0.39	
	0.92	94.78 ± 0.15	74.44 ± 0.41	91.40 ± 0.13	92.99 ± 0.05	63.27 ± 0.26	

Table 8

Quantitative evaluation of our methods with different loss weights λ . Experiments were performed based on 5 % of the labelled data with label smoothing $\alpha = 0.1$. The underlined part is the parameter chosen for our experiments.

Method	Hyperparameters weight λ	Metrics					
		AUC	Sensitivity	Specificity	Accuracy	F1	PL avg acc
FixMatch-LS-v2	0	90.30 $\pm$ 0.08	61.65 $\pm$ 0.17	89.63 $\pm$ 0.11	90.25 $\pm$ 0.12	51.57 $\pm$ 0.21	57.0
	0.01	89.60 $\pm$ 0.24	62.03 $\pm$ 0.51	89.98 $\pm$ 0.25	89.87 $\pm$ 0.19	53.15 $\pm$ 0.26	57.5
	0.1	90.20 $\pm$ 0.30	61.62 $\pm$ 0.46	89.64 $\pm$ 0.33	90.21 $\pm$ 0.18	52.38 $\pm$ 0.36	61.5
	0.5	90.52 $\pm$ 0.18	63.29 $\pm$ 0.57	89.55 $\pm$ 0.06	90.12 $\pm$ 0.19	52.96 $\pm$ 0.35	68.6
	<u>1.0</u>	<u>91.64 $\pm$ 0.04</u>	<u>64.74 $\pm$ 0.57</u>	<u>89.64 $\pm$ 0.09</u>	<u>90.77 $\pm$ 0.06</u>	<u>53.71 $\pm$ 0.33</u>	<u>71.3</u>
	5.0	91.08 $\pm$ 0.09	63.84 $\pm$ 0.33	88.48 $\pm$ 0.17	89.95 $\pm$ 0.23	52.13 $\pm$ 0.21	68.9
	10.0	90.85 $\pm$ 0.23	63.05 $\pm$ 0.66	88.09 $\pm$ 0.22	89.97 $\pm$ 0.12	52.38 $\pm$ 0.35	69.6

Table 9

Quantitative evaluation of our methods with different pseudolabels. Experiments were performed based on ISIC 2018 with 5 % of the labelled data with label smoothing $\alpha = 0.1$. The underlined part is the parameter chosen for our experiments.

Pseudolabelling				Metrics					
Index	Model	Input	Consistency	AUC	Sensitivity	Specificity	Accuracy	F1	PL avg acc
1	Student model	same		90.05 $\pm$ 0.06	62.77 $\pm$ 0.35	89.51 $\pm$ 0.08	90.39 $\pm$ 0.12	51.88 $\pm$ 0.30	65.1
2	Student model	different		90.30 $\pm$ 0.08	61.65 $\pm$ 0.17	89.63 $\pm$ 0.11	90.25 $\pm$ 0.12	51.57 $\pm$ 0.21	57.0
7	Teacher model	same		89.69 $\pm$ 0.15	61.87 $\pm$ 0.46	89.48 $\pm$ 0.17	90.19 $\pm$ 0.17	51.69 $\pm$ 0.33	62.9
8	Teacher model	different		89.82 $\pm$ 0.07	60.37 $\pm$ 0.37	89.68 $\pm$ 0.07	90.11 $\pm$ 0.06	51.00 $\pm$ 0.22	58.3
3	Student model	same	MSE + SRC	90.78 $\pm$ 0.24	60.75 $\pm$ 0.57	89.12 $\pm$ 0.16	90.60 $\pm$ 0.15	53.21 $\pm$ 0.14	72.4
4	Student model	different	MSE + SRC	91.05 $\pm$ 0.16	60.62 $\pm$ 0.34	89.61 $\pm$ 0.11	90.73 $\pm$ 0.29	53.72 $\pm$ 0.19	67.9
9	Teacher model	same	MSE + SRC	90.51 $\pm$ 0.10	58.48 $\pm$ 0.74	89.29 $\pm$ 0.07	90.61 $\pm$ 0.16	52.76 $\pm$ 0.34	67.6
10	Teacher model	different	MSE + SRC	89.87 $\pm$ 0.25	60.76 $\pm$ 0.92	89.60 $\pm$ 0.23	90.22 $\pm$ 0.22	51.61 $\pm$ 0.48	60.8
5	Student model	same	KL + SRC	91.36 $\pm$ 0.02	64.39 $\pm$ 0.60	88.97 $\pm$ 0.14	90.61 $\pm$ 0.04	53.24 $\pm$ 0.19	72.9
6	<u>Student model</u>	<u>different</u>	<u>KL + SRC</u>	<u>91.64 $\pm$ 0.04</u>	<u>64.74 $\pm$ 0.57</u>	<u>89.64 $\pm$ 0.09</u>	<u>90.77 $\pm$ 0.06</u>	53.71 $\pm$ 0.33	<u>71.3</u>
11	Teacher model	same	KL + SRC	91.15 $\pm$ 0.03	63.59 $\pm$ 0.45	89.14 $\pm$ 0.04	90.58 $\pm$ 0.07	52.96 $\pm$ 0.18	68.4
12	Teacher model	different	KL + SRC	91.03 $\pm$ 0.16	63.81 $\pm$ 0.36	89.60 $\pm$ 0.09	90.68 $\pm$ 0.17	53.37 $\pm$ 0.26	65.9

trusting 90 % for each label, which is similar to the pseudolabel threshold $T = 0.9$ ($T + \alpha = 1.0$). We speculate that this has some reference for the interpretation of the hyperparameter T . See Appendix B for more details.

D. Impact of different loss weights λ .

We investigated the effect of different hyperparameter settings in Eq. (2) on λ . λ is a Gaussian warming up function with a value range of (0,1). We investigated the effect of consistency and SRC on pseudolabelling by varying the maximum value of λ . Specifically, we adopted 7 different values on ISIC 2018 with 5 % labelled data. As shown in Table 8, the optimal choice of λ is 1.0, and increasing or decreasing this value leads to performance degradation. Note that in our framework, adjusting λ also affects the average pseudolabel accuracy. High consistency is more stringent for unlabelled data, leading to a slight decrease in the number of pseudolabels, while accuracy slightly increases.

5. Discussion

Influenced by noisy student [15], we revised the pseudolabel generation method in Fig. 1. Specifically, the teacher model is utilized to produce pseudolabels that the student model can use for training. For example, instead of using the student model in the pseudolabel generating procedure for the upper section of the branch in Fig. 1 (b), we utilized the teacher model, and the results shifted from indices 6 to 12. We also report the results of training with 5 % labelled data in Section 4.5 entitled Ablation Experiment B (indices 1 to 6) in Table 9 below for comparison.

The model with KL loss performs better than the model with MSE loss when trained with less labelled data. More pseudolabels can be generated using the teacher model, but the performance of the model does not improve. We speculate that this may be due to noisy training. First, the quality and quantity of pseudolabels varies because the teacher model is produced via EMA. Second, matching consistency regularization is missing. We suggest that the following strategies will help us improve this process. First, we can improve the smoothing mechanism of the

teacher model, e.g., [43]. Second, to obtain better pseudolabels, we can add or replace a better consistency regularization approach, e.g., [50]. Third, we can improve the label smoothing mechanism to obtain better pseudolabel quality, e.g., [37,51].

6. Conclusion

In this work, we investigate the problem of semi-supervised medical image classification to reduce the large human effort involved in labelling medical image data. We propose a new semi-supervised framework by fusing supervisory loss, consistency loss, the sample relation consistency paradigm, and pseudolabel loss. Our main contribution is the study of the effect of consistency regularization on pseudolabelling. Specifically, we reduce the impact of noisy artificial labels by introducing label smoothing, and we exploit different consistency regularization to leverage the consistency between different data augmentations of the same sample. The experiments show that our method can better exploit unlabelled data by a large margin in terms of all benchmarks containing a varying percentage of labelled data. The ablation study shows the importance of label smoothing for pseudolabelling, and consistency regularization is more effective in improving the accuracy of pseudolabels than changing the threshold value. We will investigate applying label smoothing to strong augmentation in the future, as well as matching strong augmentation with consistency regularization.

CRedit authorship contribution statement

Shaofeng Zhou: Methodology, Writing – original draft. **Shenwei Tian:** Writing – review & editing. **Long Yu:** Writing – review & editing. **Weidong Wu:** Data curation. **Dezhi Zhang:** Data curation. **Zhen Peng:** Visualization. **Zhicheng Zhou:** Supervision. **Junwen Wang:** Supervision.

Table 10

Quantitative evaluation of our methods with different α and T. Experiments were performed based on 20% of the labelled data. The underlined part is the parameter chosen for our experiments.

Method	Hyperparameter LS α	Hyperparameter threshold T	Metrics					
			AUC	Sensitivity	Specificity	Accuracy	F1	PL avg acc
FixMatch-LS	0.30	0.70	90.79 $\pm$ 0.85	75.88 $\pm$ 2.33	87.78 $\pm$ 1.07	88.00 $\pm$ 1.09	52.91 $\pm$ 1.31	7.8
	0.20	0.80	91.21 $\pm$ 0.62	76.06 $\pm$ 1.66	89.77 $\pm$ 0.84	90.63 $\pm$ 0.69	54.59 $\pm$ 1.57	15.2
	0.15	0.85	93.97 $\pm$ 0.04	74.27 $\pm$ 0.59	91.33 $\pm$ 0.11	92.72 $\pm$ 0.13	62.74 $\pm$ 0.39	90.0
	<u>0.10</u>	<u>0.90</u>	<u>94.29 $\pm$ 0.05</u>	<u>74.15 $\pm$ 0.43</u>	<u>91.68 $\pm$ 0.12</u>	<u>93.22 $\pm$ 0.09</u>	<u>63.96 $\pm$ 0.46</u>	<u>89.3</u>
	0.07	0.93	94.01 $\pm$ 0.07	73.22 $\pm$ 1.09	91.33 $\pm$ 0.08	92.99 $\pm$ 0.20	63.13 $\pm$ 0.38	92.9
	0.05	0.95	94.45 $\pm$ 0.02	73.57 $\pm$ 0.46	91.42 $\pm$ 0.11	92.93 $\pm$ 0.07	63.27 $\pm$ 0.37	89.0
	0.03	0.97	94.10 $\pm$ 0.04	73.87 $\pm$ 0.26	91.75 $\pm$ 0.03	93.21 $\pm$ 0.04	64.17 $\pm$ 0.20	89.5

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

This data is publicly available ISIC Challenge data.

Appendix A

A. Impact of label smoothing

Label smoothing systematically improves predictions of both clean and noisy samples [5]. When label smoothing $\alpha = 0.0$, 40 % of the unlabelled data are concentrated above 0.99, which corresponds to a pseudolabel accuracy of 97.3 %. The accuracy of the remaining artificial labels is less than 90 %, which is not conducive to the selection of the threshold for adding pseudolabels. When label smoothing $\alpha = 0.1$, the pseudolabelled distribution is relatively scattered and clustered around 0.9, and the accuracy of pseudolabels at 0.9 is 98.6 %. The accuracy of pseudolabels with prediction probabilities higher than 0.9 is higher than 90 % in all cases, and there are a few cases that are completely correct. The difference between the accuracies of FixMatch-LS ($\alpha = 0.1$, T = 0.9) and FixMatch-LS ($\alpha = 0.1$, T = 1.0) is not significant.

B. Different combination of α and T

We adopted 7 different combinations according to T + $\alpha = 1.0$. As shown in Table 10, when α is taken higher than 0.15, the performance degradation is severe, mainly due to excessive noise labels. When α is taken less than 0.1, the difference is not significant. The hyperparameter α directly indicates the degree of trust in the truth labels. It also reflects the relationship between categories [51]. In FixMatch-LS, α affects many components of L in Eq. (2), such as classification loss L_s , the distribution of predicted categories, the strength of consistency regularization L_c and L_{src} and the quality of pseudolabelling L_p . Considering a more general case, based on the recommended α to 0.1, we recommend T to 0.9.

References

- [1] Q. Liu, et al., Semi-supervised medical image classification with relation-driven self-ensembling model, *IEEE Trans. Med. Imaging* 39 (11) (2020) 3429–3440.
- [2] Sohn, Kihyuk, et al. Fixmatch: Simplifying semi-supervised learning with consistency and confidence, *Adv. Neural Inform. Process. Syst.* 33 (2020) 596–608.
- [3] R. Müller, S. Kornblith, G.E. Hinton, When does label smoothing help? *Adv. Neural Inf. Process. Syst.* 32 (2019).
- [4] D.-H. Lee, Pseudo-label: The simple and efficient semi-supervised learning method for deep neural networks, *Workshop on challenges in representation learning, ICML*. 3 (2) (2013) 896.
- [5] Lukasik, Michal, et al. Does label smoothing mitigate label noise?. *International Conference on Machine Learning*, PMLR, 2020: 6448–6458.
- [6] Brown, Tom, et al. Language models are few-shot learners, *Adv. Neural Inform. Process. Syst.* 33 (2020) 1877–1901.
- [7] Hu, Ronghang, Amanpreet Singh, Transformer is all you need: Multimodal multitask learning with a unified transformer, *arXiv preprint arXiv:2102.10772*, (2021).
- [8] Hady, Mohamed Farouk Abdel, F. Schwenker, Semi-Supervised Learning, *J. Royal Statist. Soc. B* 172.2 (2006) 530–530.
- [9] P. Bachman, O. Alsharif, D. Precup, Learning with Pseudo-Ensembles, In: *Advances in neural information processing systems*, pages 3365–3373, 2014.
- [10] S. Laine, T. Aila, Temporal Ensembling for Semi-Supervised Learning, *International Conference on Learning Representations*, (2017). 10.48550/arXiv.1610.02242.
- [11] F. Perez, et al., Data augmentation for skin lesion analysis, in: *OR 2.0 Context-Aware Operating Theaters, Computer Assisted Robotic Endoscopy, Clinical Image-Based Procedures, and Skin Image Analysis*, Springer, Cham, 2018, pp. 303–311.
- [12] Tarvainen, Antti, and Harri Valpola, Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results, *Adv. Neural Inform. Process. Syst.* 30 (2017). 10.48550/arXiv.1703.01780.
- [13] T. Miyato, et al., Virtual adversarial training: a regularization method for supervised and semi-supervised learning, *IEEE Trans. Pattern Anal. Mach. Intell.* 41 (8) (2018) 1979–1993.
- [14] Xie, Qizhe, et al. Unsupervised data augmentation for consistency training, *Adv. Neural Inform. Process. Syst.*, 33 (2020) 6256–6268.
- [15] Q. Xie, M.-T. Luong, E. Hovy, et al. Le. Self-training with noisy student improves imagenet classification, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10687–10698, (2020).
- [16] D. McClosky, E. Charniak, M. Johnson, Effective self-training for parsing, in: *Proceedings of the main conference on human language technology conference of the North American Chapter of the Association of Computational Linguistics. Association for Computational Linguistics*, (2006).
- [17] C. Rosenberg, M. Hebert, H. Schneiderman, Semi-supervised self-training of object detection models, in: *Proceedings of the Seventh IEEE Workshops on Application of Computer Vision*, vol. 1, pp. 29–36, (2005).
- [18] Berthelot D, Carlini N, Goodfellow I, et al. Mixmatch: A holistic approach to semi-supervised learning, in: *Advances in Neural Information Processing Systems*, pages 5050–5060, 2019.
- [19] D. Berthelot, N. Carlini, E.D. Cubuk, et al. Remixmatch: Semi-supervised learning with distribution alignment and augmentation anchoring, *arXiv preprint arXiv:1911.09785*, (2019).
- [20] Oliver, Avital, et al. Realistic evaluation of semi-supervised learning algorithms, *Int. Conf. Learn. Represent.* 2018, 1–15.

- [21] Szegedy, Christian, et al. Rethinking the inception architecture for computer vision, *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, 2818–2826.
- [22] Ashish Vaswani, et al. Attention is all you need, in: *Advances in neural information processing systems*, pages 5998–6008, (2017).
- [23] J. Chorowski, N. Jaitly, Towards better decoding and language model integration in sequence to sequence models, *Proc. Interspeech 2017* (2017) 523–527.
- [24] P. Tschandl, C. Rosendahl, H. Kittler, The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions, *Sci. Data* 5 (1) (2018) 1–9.
- [25] Hongyi Zhang, Moustapha Cisse, Yann N. Dauphin, et al. mixup: Beyond empirical risk minimization, in: *International Conference on Learning Representations*, (2018). arXiv preprint arXiv:1710.09412, (2017).
- [26] Wei, Chen, et al. Crest: A class-rebalancing self-training framework for imbalanced semi-supervised learning, *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2021, 10857–10866.
- [27] V. Cheplygina, M. de Bruijne, J.P.W. Pluim, Not-so-supervised: a survey of semi-supervised, multi-instance, and transfer learning in medical image analysis, *Med. Image Anal.* 54 (2019) 280–296.
- [28] Kitada, Shunsuke, and Hitoshi Iyatomi, Skin lesion classification with ensemble of squeeze-and-excitation networks and semi-supervised learning. arXiv preprint arXiv:1809.02568 (2018).
- [29] Li, Xiaomeng, et al. Semi-supervised skin lesion segmentation via transformation consistent self-ensembling model, arXiv preprint arXiv:1808.03887 (2018).
- [30] J. Zhang, et al., Attention residual learning for skin lesion classification, *IEEE Trans. Med. Imaging* 38 (9) (2019) 2092–2103.
- [31] X. Shi, et al., An active learning approach for reducing annotation cost in skin lesion analysis, in: *international Workshop on Machine Learning in Medical Imaging*, Springer, Cham, 2019, pp. 628–636.
- [32] F. Liu, et al., Self-supervised mean teacher for semi-supervised chest x-ray classification, in: *international Workshop on Machine Learning in Medical Imaging*, Springer, Cham, 2021, pp. 426–436.
- [33] Zoph, V. Vasudevan, J. Shlens, et al. Learning transferable architectures for scalable image recognition, in: *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 8697–8710, (2018).
- [34] Rizve, Mamshad Nayeem, et al. In defense of pseudo-labeling: An uncertainty-aware pseudo-label selection framework for semi-supervised learning, arXiv preprint arXiv:2101.06329 (2021).
- [35] Combalia, Marc, et al. Bcn20000: Dermoscopic lesions in the wild. arXiv preprint arXiv:1908.02288 (2019).
- [36] Shi, Weiwei, et al. Transductive semi-supervised deep learning using min-max features, *Proceedings of the European Conference on Computer Vision (ECCV)*. 2018 299–315.
- [37] Maher, Mohamed, and Meelis Kull. Instance-based Label Smoothing For Better Calibrated Classification Networks, *2021 20th IEEE International Conference on Machine Learning and Applications (ICMLA)*. IEEE, 2021, 746–753.
- [38] Z. Qin, et al., A GAN-based image synthesis method for skin lesion classification, *Comput. Methods Programs Biomed.* 195 (2020) 105568.
- [39] Y. Grandvalet, Y. Bengio, Semi-supervised Learning by Entropy Minimization, *Advances in Neural Information Processing Systems* 17 [Neural Information Processing Systems, NIPS 2004, December 13–18, 2004, Vancouver, British Columbia, Canada] MIT Press, 2005.
- [40] Cubuk, Ekin D., et al. Randaugment: Practical automated data augmentation with a reduced search space, *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*. (2020). doi.org/10.48550/arXiv.1909.13719.
- [41] A. Diaz-Pinto, et al., Retinal image synthesis and semi-supervised learning for glaucoma assessment, *IEEE Trans. Med. Imaging* 38 (9) (2019) 2211–2218.
- [42] W. Bai, et al., Semi-supervised learning for network-based cardiac MR image segmentation, in: *International Conference on Medical Image Computing and Computer-Assisted Intervention*, Springer, Cham, 2017, pp. 253–260.
- [43] Huang, Tengting, et al. Spatial Ensemble: a Novel Model Smoothing Mechanism for Student-Teacher Framework, *Adv. Neural Inform. Process. Syst.*, 34 (2021). http://doi.org/10.48550/arXiv.2110.01253.
- [44] Xiong, Yunsheng, et al. WRMatch: Improving FixMatch with Weighted Nuclear-Norm Regularization for Few-Shot Remote Sensing Scene Classification, *IEEE Trans. Geosci. Remote Sens.* (2021). http://doi.org/10.1109/TGRS.2021.3121765.
- [45] Zhang, Bowen, et al. Flexmatch: Boosting semi-supervised learning with curriculum pseudo labeling, *Adv. Neural Inform. Process. Syst.*, 34 (2021) 18408–18419.
- [46] Wang, Can, et al. Pseudo-Labeled Auto-Curriculum Learning for Semi-Supervised Keypoint Localization, arXiv preprint arXiv:2201.08613 (2022).
- [47] Li, Junnan, Caiming Xiong, et al. Comatch: Semi-supervised learning with contrastive graph regularization, *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, 9475–9484.
- [48] Liu, Fengbei, et al. ACPL: Anti-Curriculum Pseudo-Labeling for Semi-Supervised Medical Image Classification, *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, 20697–20706.
- [49] Kim, Taehyeon, et al. Comparing kullback-leibler divergence and mean squared error loss in knowledge distillation, arXiv preprint arXiv:2105.08919 (2021).
- [50] Han, Tao, Wei-Wei Tu, and Yu-Feng Li. Explanation consistency training: Facilitating consistency-based semi-supervised learning with interpretability, *Proceedings of the AAAI conference on artificial intelligence*. Vol. 35. 2021.
- [51] Lienen, Julian, and Eyke Hüllermeier. From label smoothing to label relaxation, *Proceedings of the 35th AAAI Conference on Artificial Intelligence*, AAAI, Online. 2021.
- [52] Zhengfeng Lai, et al. Smoothed Adaptive Weighting for Imbalanced Semi-Supervised Learning: Improve Reliability Against Unknown Distribution Data, *Proceedings of the 39th International Conference on Machine Learning*, PMLR 162 (2022) 11828–11843.